

Supporting Information

for

Deducing Reaction Mechanism: A Guide for Students, Researchers, and Instructors

Simon J. Meek, Catherine L. Pitman, and Alexander J. M. Miller

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I. Rules of Thumb for Mechanistic Studies

- Increasing the reaction temperature by 10 degrees will result in the rate constant approximately doubling (see Arrhenius equation). *Caution:* despite being widely used, this is valid for very few reactions — single-step reactions with $\Delta G^\ddagger$ (at 25 °C) in the range of 10-15 kcal·mol⁻¹.
- A difference in barrier heights of only $\Delta\Delta G^\ddagger = 2.3$ kcal·mol⁻¹ results in 50:1 kinetic selectivity for the product formed by the lower energy pathway for a reaction in which a single reactant can form two different products *at room temperature* (that would be good for a 98:2% er).
- For an equilibrium reaction *at 25 °C* involving equal numbers of reactants and products (unitless K_{eq}), a difference in the free energy of reaction of $\Delta G^\circ = 1.36$ kcal·mol⁻¹ changes K_{eq} by a factor of 10. A reaction that favors products by only 2 kcal·mol⁻¹ will contain 97% product.
- The rate constant of a unimolecular reaction with a reasonable rate (half-life ~2 hours at room temperature) is on the order of 10⁻⁴ s⁻¹, with $\Delta G^\ddagger \sim 23$ kcal·mol⁻¹ (at 25 °C).
- The fastest possible rate of a unimolecular reaction (for the case of $\Delta G^\ddagger = 0$) is $k_B T/h$ (according to the Eyring equation). At 25 °C this corresponds to about 10¹³ s⁻¹.
- The fastest possible rate of a bimolecular reaction is limited by the rate at which the diffusion brings the two reactants together. This varies with temperature and solvent viscosity, but is generally in the range of 10⁹ – 10¹⁰ M⁻¹·s⁻¹.

- The entropy of reaction, ΔS° , is usually close to $0 \text{ cal}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ for reactions in which one reactant molecule is converted to one product molecule. The entropy of reaction (ΔS°) is usually large and negative ($\sim -30 \text{ cal}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$) for reactions in which two reactant molecules are converted to a single product. The entropy of reaction (ΔS°) is usually large and positive ($\sim +30 \text{ cal}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$) for reaction in which a single reactant is converted to two products.
- The entropy of activation ($\Delta S^\ddagger$) is usually large and negative ($\sim -25 \text{ cal}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$) for reactions in which two reactant molecules are converted to a single product. The entropy of activation ($\Delta S^\ddagger$) is usually small and positive ($\sim +7 \text{ cal}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$) for reaction in which a single reactant is converted to two products. The entropy of activation of unimolecular reactions depends on the degree of order in the transition state and can vary widely.

II. Bibliography of Selected Mechanistic Studies

The following publications were selected as appropriate for use as case studies in the classroom. Many of these studies have been used in the reaction mechanisms course at UNC, either as topics of classroom discussion or in problem session and exam problems.

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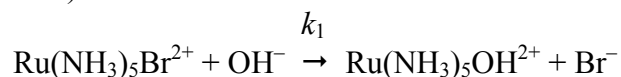
III. Example Group Problems and Exam Assessments

The example questions provided below are based on literature examples, although the data are often altered slightly for teaching purposes. The questions used in problem solving sessions focus in depth on a single recently introduced topic. Some of these questions are worked in class and discussed part-by-part, while others are left as out-of-class exercises. The exam questions combine several concepts applied to a literature example. The questions below are arranged with several in-class questions on specific topics, followed by multi-part exam questions that draw from several of the covered topics.

A. Questions used in weekly in-class group problem solving sessions.

Topic: Rate Laws

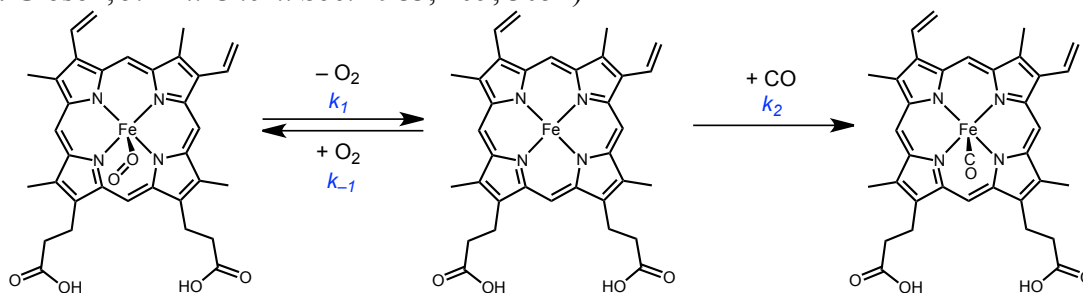
Question: Consider a substitution reaction (C. H. Langford, W. R. Muir, *J. Am. Chem. Soc.* **1967**, 89, 3141–3144.).



- Derive the rate law, assuming an elementary reaction.
- Consider the case where $[\text{OH}^-]$ is more than 10 times $[\text{RuBr}^{2+}]$. How would you determine the pseudo-first order rate constant, k_{obs} ?
- Consider the case where $[\text{OH}^-] = [\text{RuBr}]$, with $[\text{OH}^-]$ dictated by a buffer. How would you determine the pseudo-first order rate constant, k_{obs} ?
- As you know, the assumption in part A is not necessarily valid. How could you determine the reaction order in $[\text{OH}^-]$. How could you obtain the true rate constant, k_1 ?

Topic: Steady-state approximation

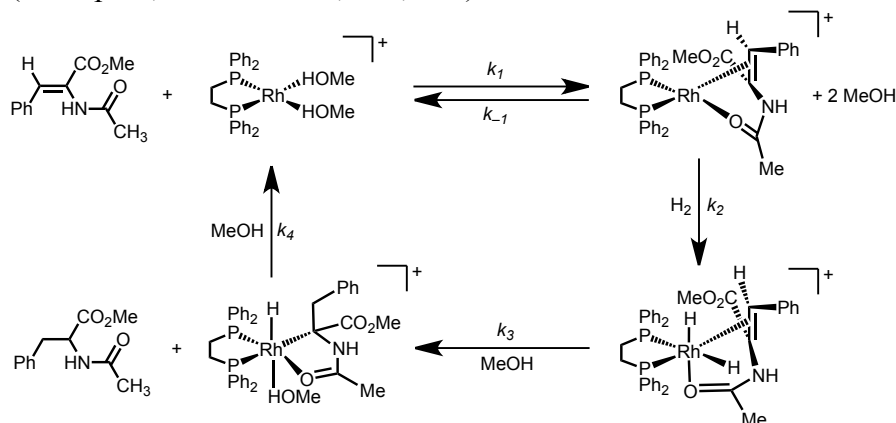
Question: Consider the substitution of O_2 by CO in Heme b (iron porphyrin complex) drawn below. (J. P. Collman, J. I. Brauman, B. L. Iverson, J. L. Sessler, R. M. Morris, Q. H. Gibson, *J. Am. Chem. Soc.* **1983**, *105*, 3052)



- A. Derive the rate law, applying the steady-state approximation. Work through the whole derivation and show your work. *Hint*: start by re-writing the scheme above using a convenient shorthand notation.
- B. What do you expect the order in O_2 to be? How would high O_2 pressures impact the reaction?
- C. What do you expect the order in CO to be? How would high CO pressure impact the reaction? Under the high O_2 pressure conditions of Part B (and low CO pressure), will the order in CO change?
- D. Under pseudo-first order conditions (large excess of CO and O_2), the rate law simplifies to the typical steady-state case. Using the distributed spreadsheet, try to simulate the effect of the conditions of Parts B and C on the distribution of species. Are the conditions of Parts B and C adequately modeled by the steady-state approximation?
- E. Based on your answers above, and realizing that Heme is the O_2 carrier in red blood cells, how would you treat a victim of CO poisoning?

Topic: Catalysis

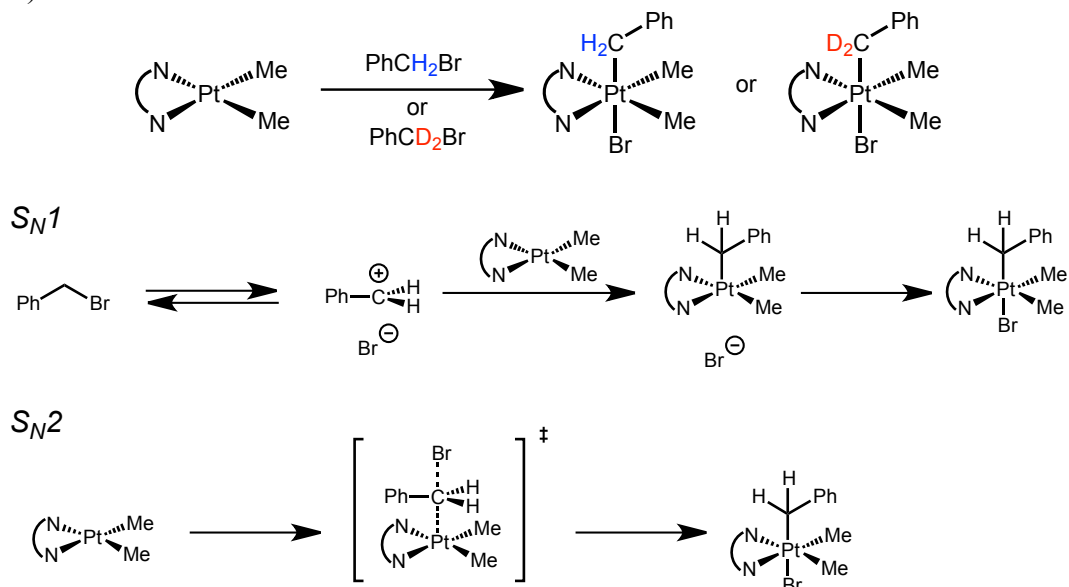
Question (J. Halpern, *Science* **1982**, 217, 401)



- Consider the achiral hydrogenation reaction in the Scheme above. The reaction solvent is methanol and Step 3 and Step 4 are fast (k_3 and k_4 are large).
 - Label each step with the type of reaction.
 - Which are elementary steps?
- Re-write the catalytic cycle as a series of equations, using abbreviations for the metal complexes.
- Consider Step 1. The rate constants were determined by rapid-mixing stopped-flow experiments: $k_1 = 1.4 \times 10^4$ and $k_{-1} = 5.2 \times 10^{-1}$.
 - What are the units of k_1 and k_{-1} ?
 - Write an equation for K_{eq} in terms of the concentration of reagents.
 - What is the value of K_{eq} ?
 - Suggest an experiment to verify the value derived from k_1 and k_{-1} .
- Now let's consider the full catalytic cycle.
 - Based on the rate constants and/or equilibrium constant, which form of the Michaelis-Menten equation is likely to be applicable for this catalytic system? Write the appropriate Michaelis-Menten rate law.
 - What substrate is $[S]$ in the Michaelis-Menten equation?
 - Explain how you could determine k_2 graphically. How would you know that your equation was correct? What is the significance of k_2 ?

Topic: Kinetic Isotope Effects

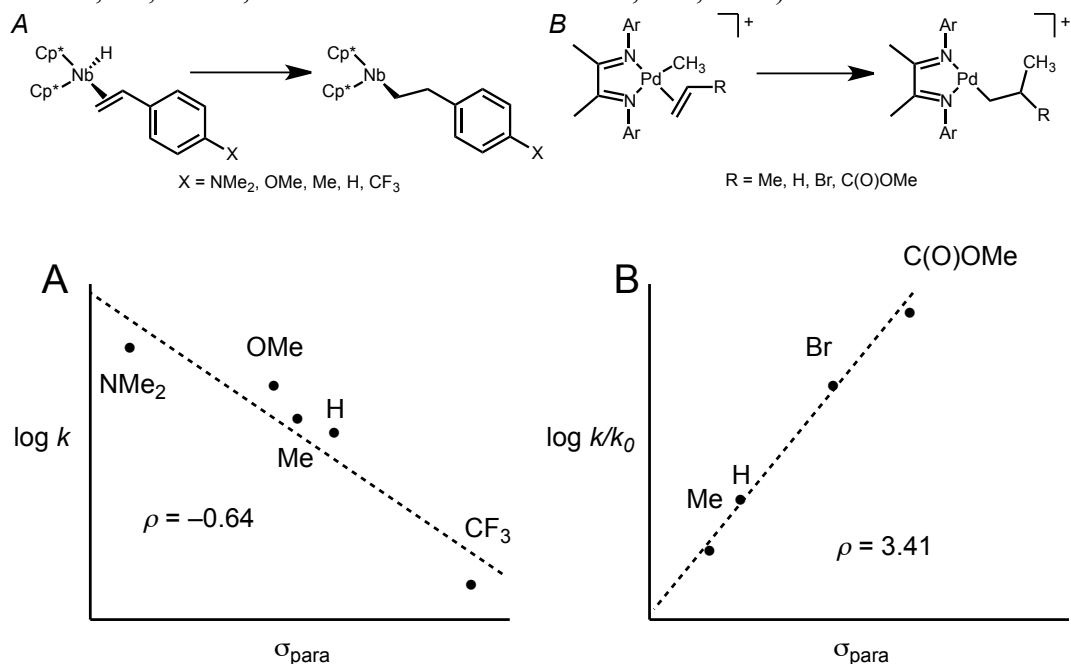
Question: Puddephatt et al. recently used kinetic isotope effects to determine the mechanism of the oxidative addition of PhCH_2Br to dimethylplatinum(II) complexes. The rates of reaction of $\text{PtMe}_2(\text{bpy})$ with PhCH_2Br and PhCD_2Br were independently measured giving $k_{\text{H}}/k_{\text{D}} = 0.97(1)$ (acetone) and $k_{\text{H}}/k_{\text{D}} = 1.25(1)$ (benzene) (Aseman, M. D.; Rashidi, M.; Nabavizadeh, S. M.; Puddephatt, R. J., *Organometallics*, **2013**, 32, 2593).



- What type of kinetic isotope effect is observed in each solvent?
- Draw a reaction coordinate diagram illustrating the origin of the isotope effect in acetone.
- The dimethylplatinum complex could be the nucleophile in either an S_N1 or S_N2 reaction. Based on the KIE in acetone, which mechanism do you predict is operating?
- Would you expect the reaction to be faster in acetone or in benzene? Why?

Topic: Hammett plots and linear free energy relationships.

Question: The alkene insertion reactions drawn below were studied independently by Bercaw and Brookhart. Hammett plots were obtained by measuring the reaction rate constant for alkenes with a variety of substituents, R. (Burger, B. J.; Santarsiero, B. D.; Trimmer, M. S.; Bercaw, J. E. *J. Am. Chem. Soc.* **1988**, *110*, 3134 and Rix, F. C.; Brookhart, M.; White, P. S. *J. Am. Chem. Soc.* **1996**, *118*, 2436).

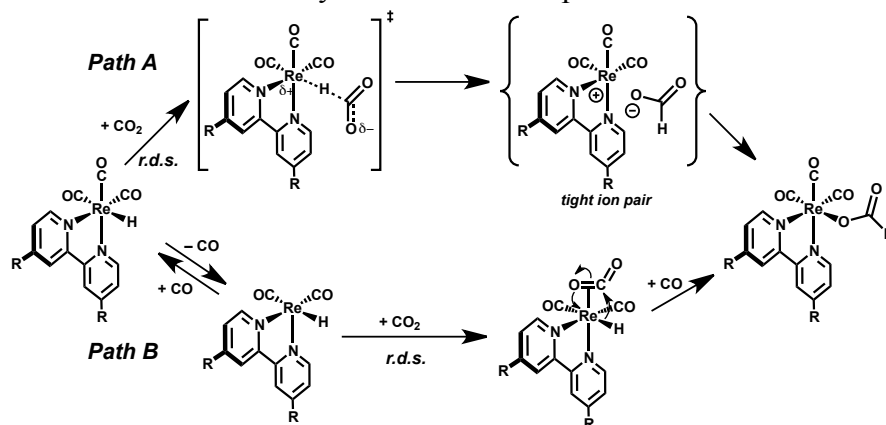


- Draw a transition state for Reaction A. Label any developing charges.
- Explain how your transition state is consistent with the Hammett plot for Reaction A. (If it is inconsistent, draw a different TS!) Draw a reaction coordinate diagram showing how the transition state energy changes.
- Why is the Hammett plot for the Reaction B is different? Draw a reaction coordinate diagram to support your answer. *Hint:* the authors attributed the free-energy relationship to a ground state effect.
- For Reaction B, there has been debate about why ethylene inserts into a Pd-CH₃ faster than propylene. Is it a steric or electronic effect?

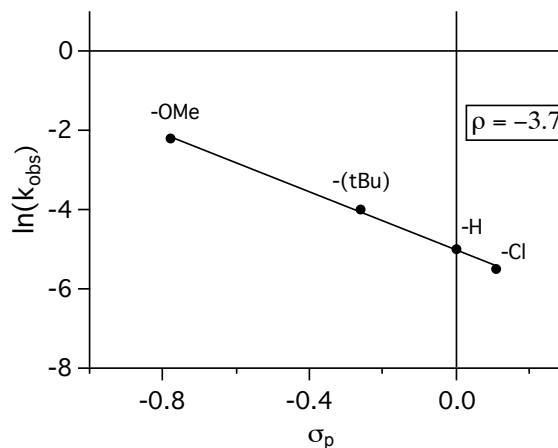
B. Questions used in in-class exams.

Exam Question 1 (Sullivan, B. P.; Meyer, T. J. *Organometallics* **1986**, 5, 1500)

Prof. T. J. Meyer considered two mechanisms for CO₂ reduction by a Re–H. Kinetic analysis revealed a second-order rate law, $-\text{d}[\text{ReH}]/\text{dt} = k[\text{ReH}][\text{CO}_2]$. The independently measured rates of CO₂ reduction by Re–H and Re–D provided $k_{\text{H}}/k_{\text{D}} = 0.55$.



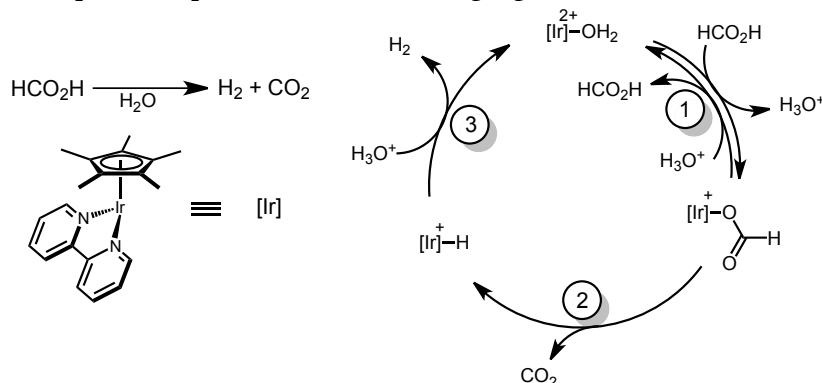
- Write the expected rate laws for Path A and Path B. (No intermediates were observed.)
- Which pathway is indicated by the kinetic isotope effect? Explain briefly.
- Why is an *inverse* kinetic isotope effect observed? Explain briefly.
- Substituted bipyridines (R = MeO, *t*Bu, H, Cl) allowed construction of the Hammett plot below. Which pathway is indicated by this data? Explain briefly.



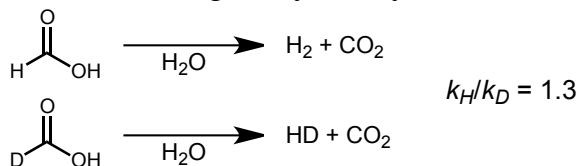
- Recommend an additional experiment that could further strengthen your mechanistic proposal. Briefly explain what you would expect to observe.

Exam Question 2 (Inspired by: Fukuzumi, S.; Kobayashi, T.; Suenobu, T. *J. Am. Chem. Soc.* **2010**, *132*, 1496 and Himeda, Y. *Green Chem.* **2009**, *11*, 2018)

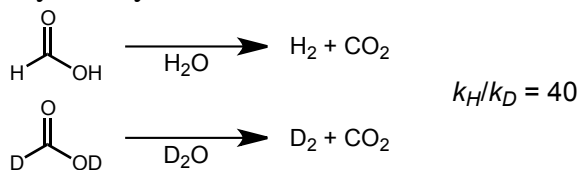
The decomposition of formic acid to H_2 and CO_2 is downhill ($\Delta G^\circ = -8 \text{ kcal}\cdot\text{mol}^{-1}$), but kinetically slow in the absence of a catalyst. The Ir complex below was recently shown to rapidly produce H_2 and CO_2 from formic acid. A proposed mechanism is shown below.



- Write a rate law using the pre-equilibrium approximation, assuming that step 2 is rate-determining. The pre-equilibrium approximation is applied because the Ir formate complex was observed in small amounts during the reaction.
- Write the rate law in the Michaelis-Menten form.
- The decomposition of formic acid-*d* in water proceeded with a KIE of 1.3 (from separate initial rates experiments). Is this consistent with the proposed mechanism where step 2 is rate-determining? Why or why not?

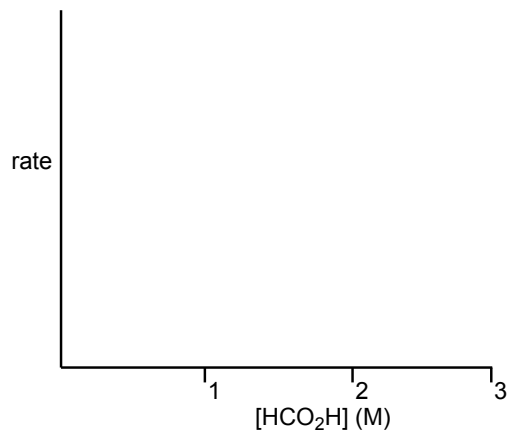
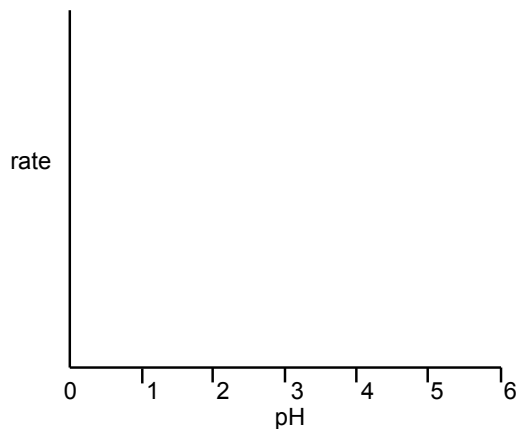


- The decomposition of formic acid-*d*₂ in deuterium oxide solvent proceeded with a KIE of 40. Is this consistent with the proposed mechanism where step 2 is rate-determining? Why or why not?



E. What step do you think is rate-determining, based on the above data?

F. Based on your answer to part E, predict the rate dependence on (a) pH and (b) concentration formic acid by filling in the graphs below. Briefly explain your answer.



IV. Good Fits and Bad Fits.

One reason that linear fits are so popular as graphical tools is the ability of the human eye to quickly assess the goodness of fit. The R^2 value is a primary quantitative measure of a successful fit, but it should not be the only consideration. In the example below, two mock data sets are graphed for the decay of some species A and another species B. In Figure S1, both plots A and B have R^2 values above 0.9, so one might conclude that the data are both fit well by a line, and A and B decay by the same mechanism.

Upon visual inspection, however, it is clear that plot A in Figure S1 shows only random scatter on either side of the linear fit, whereas plot B in Figure S1 shows systematic curvature. Such systematic curvature is cause for concern, even when high R^2 values are obtained for a fit. One should always consider other pathways, but when data curvature is observed, it is imperative to examine other possibilities.

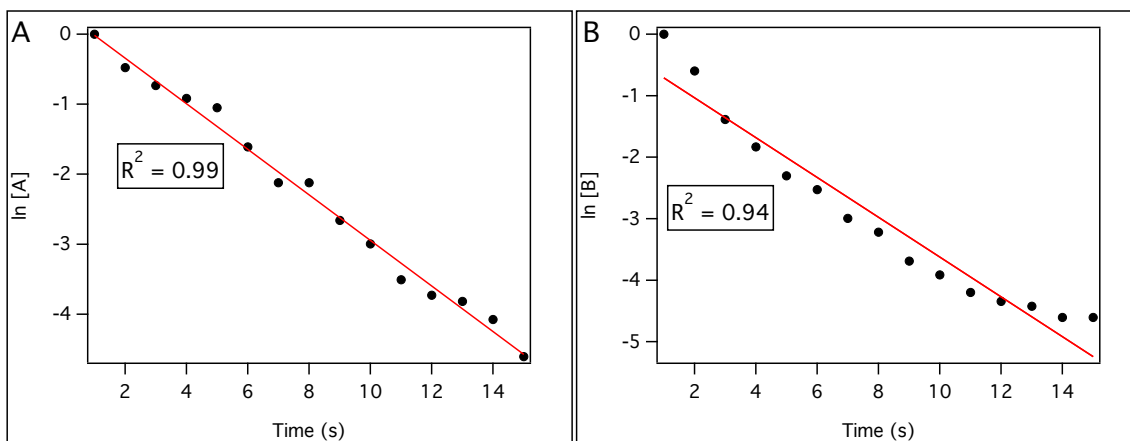


Figure S1. First order decay plots using mock data for the theoretical decay of a species A (plot A) and a species B (plot B). The random scatter in A and the high R^2 are indicative of a good linear fit, whereas the lower R^2 and the systematic curvature in B are indicative of a poor linear fit.

As can be seen in Figure S2, the decay of B is much better fit by a plot of $1/[B]^{1/2}$ vs. time, corresponding to an order of 3/2. The concerning curvature is no longer present, and random scatter is observed. The residual of a fit can provide similar checks of randomness or scatter in a fit. As shown in Figure S3, the residual from the fit in Figure S1B shows a systematic curvature, whereas the fit from Figure S2 is random.

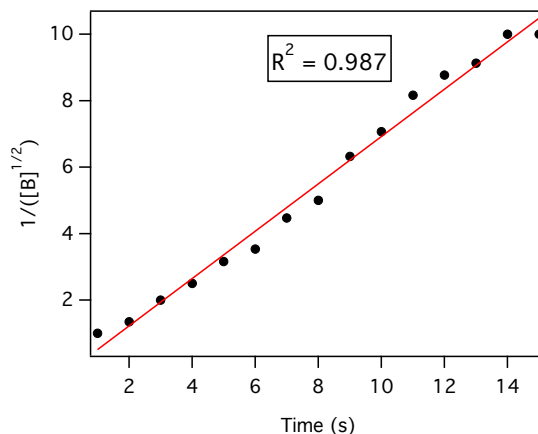


Figure S2. Three-halves order plot of the mock data for the theoretical decay reaction of B, illustrating a good linear fit.

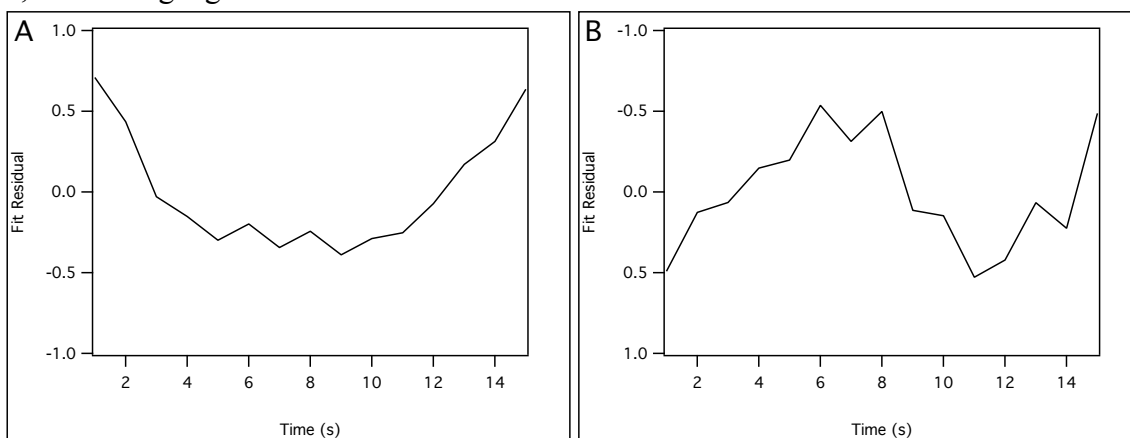


Figure S3. Comparing the linear fit residuals from a first order plot (A) and a three-halves order plot (B). The curvature in A is cause for concern, whereas the random nature of the residual in B is indicative of a better fit.

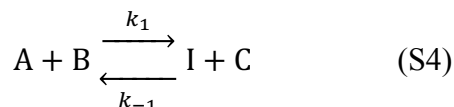
V. Steady-State Approximations for Complex Reactions.

In the main text, the approximations for multi-step reactions were discussed only in the context of simple unimolecular reactions. Higher-order reactions can also be treated according to these approximations. The same derivations can be performed, but it is often easier to deduce the rate law by inspection. This is done by inserting a concentration term next to the rate constant relevant to the reaction of that reactant. Alternatively, one can consider the rate to be equal to the *product* of all the forward reactions (k_1 and k_2), divided by the *sum* of all the reactions that consume [I] (k_{-1} and k_2). For the example of the bimolecular reaction in Equations S1 and S2, the rate law would take the form of Equation S3: reactant B will appear anywhere that k_2 appears in the rate law for the simple case, with $k_{obs} = k_1k_2/(k_{-1}+k_2)$.



$$rate = \frac{-d[A]}{dt} = \frac{k_1[A]k_2[B]}{k_{-1}+k_2[B]} \quad (S3)$$

For the more complex reaction series of Equations S4 and S5, the rate law can again be deduced by inspection. In this case, species C is formed in the first step, and is required for the reverse reaction k_{-1} . Thus, a [C] term appears on the denominator, and one would expect that including extra C in the reaction mixture would slow down the reaction.



$$rate = \frac{-d[A]}{dt} = \frac{k_1[A][B]k_2[D]}{k_{-1}[C] + k_2[D]} \quad (S6)$$

Saturation behavior is a hallmark of a hidden intermediate in complex reactions. Saturation kinetics arise from the presence of a concentration term of a reactant on both the top and the bottom of the rate law of a multi-step reaction. If we again consider the reaction scheme of Equations S1 and S2, we have the rate law of Equation S7:

$$rate = \frac{d[P]}{dt} = \frac{k_1[A]k_2[B]}{k_{-1} + k_2[B]} \quad (S7)$$

At very low [B], the denominator is dominated by k_{-1} ($k_{-1} \gg k_2[B]$), so Equation S7 simplifies to Equation S8, which would be first order in [B], with a plot of rate vs. [B] giving a linear relationship.

$$rate = \frac{d[P]}{dt} = \frac{k_1[A]k_2[B]}{k_{-1}} \quad (S8)$$

On the other hand, when [B] gets large, the denominator is dominated by $k_2[B]$ ($k_2[B] \gg k_{-1}$), leading to the new rate law in Equation S9, which further simplifies to the rate law of Equation S10. Equation S10 is clearly zero order in [B], so a plot of rate vs. [B] would show no dependence (horizontal line).

$$rate = \frac{d[P]}{dt} = \frac{k_1[A]k_2[B]}{k_2[B]} \quad (S9)$$

$$rate = \frac{d[P]}{dt} = k_1[A] \quad (S10)$$

Based on these two limiting scenarios, saturation kinetics are indicated when a plot of rate vs. concentration exhibits a linear portion at low reactant concentration before showing less and less dependence on reactant concentration, ultimately reaching a

horizontal portion of zero order dependence. Michaelis-Menten plots are a prime example of saturation behavior, as shown in Figure 4 in the main text.